

Web Browsing, Search, Upload and Download — Detailed Lecture Notes

JKSSB Computer Awareness

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36.1 The browser, tabs, bookmarks, and history

A **web browser** is the software used to open and view web pages on the internet. Common examples are Google Chrome, Mozilla Firefox, and Microsoft Edge. The browser sends a request for a page and then displays the returned text, images, and links. The address typed to reach a page is the **URL** — Uniform Resource Locator — and it is typed in the **address bar** at the top of the browser window.

Definition (Web browser). *Software that requests web pages from servers and displays them to the user.*

Term	Meaning in this lesson
Web browser	Software (Chrome, Firefox, Edge) that opens and displays web pages.
Tab	One open page inside the browser window; many tabs can stay open together.
Bookmark	A saved shortcut to a page for quick return without retyping the address.
Browsing history	The date-wise list of pages visited earlier, kept by the browser.
URL	Uniform Resource Locator: the address of a page, typed in the address bar.
Search engine	A website (Google, Bing) that finds pages matching typed words.

Each open page lives in its own **tab**, so several pages can stay open in one window. A new tab is opened with **Ctrl+T**. A **bookmark** is a saved shortcut to a page: instead of typing the address again, the saved bookmark is clicked. It is created with **Ctrl+D**. **Browsing history** is the list of pages visited earlier, stored by date, and it is opened with **Ctrl+H**. The distinction is simple: history answers “where was I”, while a bookmark answers “where do I want to return”.

Example 36.1.1. *While preparing a report on the office desktop, the clerk keeps the office portal open in one tab and the search results in a second tab, and bookmarks the portal page so it can be opened directly the next morning. The history list that evening shows both pages in order.*

36.2 How a search engine works

A **search engine** is a website, such as Google or Bing, that finds pages matching the words typed by the user. It works in three ordered stages: crawling, indexing, and ranking.

Definition (Crawling). *The collection step: automatic programs called crawlers or spiders visit pages and follow their links to discover new and changed pages.*

Definition (Indexing). *The storage step: the crawled pages are kept in a large index, like the index at the back of a book, so matches can be found quickly.*

Definition (Ranking). *The ordering step: when words are typed, matching pages are picked from the index and ordered by relevance on the search results page.*

Crawling comes first because without it the engine would never learn that a new page exists. Indexing comes second: the collected pages are stored in the index before any search is typed. Ranking comes last: only after the user types words does the engine select matches from that ready index and order them. The anchor to remember is the order itself: **crawl, then index, then rank**.

36.3 Sponsored results and organic results

The matches chosen by relevance are called **organic results**. In addition, the results page shows **sponsored results**: paid placements that appear because an advertiser paid for them, not because they ranked highest on relevance. A sponsored result always carries a visible **Ad** or **Sponsored** label. The practical consequence is direct: being listed first on the page does not always mean being the most relevant match, since the top position may be a paid placement.

Example 36.3.1. *A search for a printer shows two blocks: the first carries a small “Ad” label and links to a seller who paid for the position, while the entries below it are the organic results ordered by relevance. The topmost entry is therefore not necessarily the best match.*

36.4 Cookies, cache, and history

These three stores are the most confused triple in this lesson. Each stores a different thing, and each is stored by a different party.

Definition (Cookie). *A small file a website stores on the computer to remember that visitor: login state, cart items, and preferences such as language.*

Definition (Browser cache). *Copies of page parts (images, styles, scripts) kept by the browser on the computer so repeat visits load faster.*

Definition (Browsing history). *The list of visited page addresses, kept by the browser by date.*

Item	What it stores	Who stores it
Cookie	Visitor data: login, cart, choices	The website
Cache	Copies of page files: images, styles	The browser
History	List of visited page addresses	The browser

Cookies are set by websites and read back on the next visit, which is why a user stays logged in. The cache is managed by the browser: on the next visit the saved copies load from the disk instead of being fetched again, so the page opens faster. Clearing them has different effects. Deleting **cookies** signs the user out of sites and clears saved preferences. Clearing the **cache** does not sign the user out; it only makes the next visit slightly slower. Clearing **history** removes the record of visited addresses but changes neither logins nor saved page copies.

36.5 Downloading and uploading

Downloading means copying a file from the internet to the user's computer: saving a form PDF from the office portal, or saving a photo from a web page. **Uploading** means sending a file from the user's computer to the internet: attaching a document to an email, or submitting a scanned form on a portal. The direction is always judged from the user's machine: download moves data **down** to the machine, upload moves data **up** from the machine to a server.

Definition (Downloading). *Copying a file from the internet (a server) to the user's computer.*

Definition (Uploading). *Sending a file from the user's computer to the internet (a server).*

Point	Download	Upload
Direction	Server to computer	Computer to server
Typical use	Opening pages, saving files	Mail attachments, form submission
Typical speed	Faster	Slower

Home and office connections usually provide a faster download speed than upload speed, because receiving pages and files is the more common need. A downloaded file, once saved, opens without needing the internet; an upload needs the connection at the moment of sending.

36.6 File conversion

File conversion means changing a file from one format to another while keeping the same content. The common office need is converting a Word document (**.docx**) to **PDF** before submitting it on a portal. Conversion changes the **format**, not the transfer direction: a converted file must still be uploaded or downloaded separately. The **extension** (such as .pdf, .docx, .jpg) is the name tag that says which format the file is stored in, while the **format** (such as PDF, PNG) is the storage method itself. The two words must not be conflated: the extension names the format; it is not the format.

Example 36.6.1. *The clerk finishes the report in Word, converts the .docx file to PDF, and then uploads the PDF on the portal. The conversion changed the format; the upload changed the location. If the portal instead needed a photograph, a .JPG image would be converted to the needed picture format the same way.*

36.7 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. The search order is **crawl**, then **index**, then **rank**. The index is built before the search is typed.
2. **Sponsored** means a paid placement with an Ad label; **organic** means ranked by relevance. The top of the page is not always the best match.
3. **Cookie** (website; visitor data) vs **cache** (browser; page copies) vs **history** (browser; visited list). Do not swap the three.

4. **Download** means internet to computer; **upload** means computer to internet. Judge the direction from the user's machine.
5. **Conversion** changes the format; **transfer** changes the location. The extension (.png) names the format (PNG); it is not the format itself.

Exercises

1. List the three stages of a search engine in order and say what each stage does.
2. Distinguish a sponsored result from an organic result.
3. Distinguish a cookie from the browser cache and from browsing history.
4. Explain the difference between downloading and uploading, giving one office example of each.
5. What does file conversion change, and how is the extension related to the format?

Solutions

1. Crawling (crawlers visit pages and follow links to collect them), then indexing (pages stored in a large index), then ranking (matches picked from the index and ordered by relevance after words are typed).
2. A sponsored result is a paid placement carrying an Ad or Sponsored label; an organic result is a normal match chosen and ordered by relevance.
3. A cookie is visitor data (login, cart, choices) stored by the website; the cache is copies of page files (images, styles) stored by the browser for faster repeat visits; history is the browser's date-wise list of visited addresses.
4. Downloading copies a file from the internet to the computer, such as saving a form PDF from the portal; uploading sends a file from the computer to the internet, such as attaching a document to an email.
5. Conversion changes the format while keeping the content (for example .docx to PDF). The extension (.pdf, .docx) is the name tag for the format; the format is the storage method itself.